Read Mapping

1. True or False: The most efficient way to map reads to a reference genome is by assembling the reads into a genome, then comparing this genome with the reference.

    True

    False

2. True or False: The most efficient way to map reads to a reference genome is by using global alignment to align each read to the reference genome.

    True

    False

3. What is the runtime of BruteForcePatternMatching?

    O(|*Text*|*|*Patterns*|)

    O(|*Text*|^2)

    O(|*Text*|)

    O(|*Patterns*|)

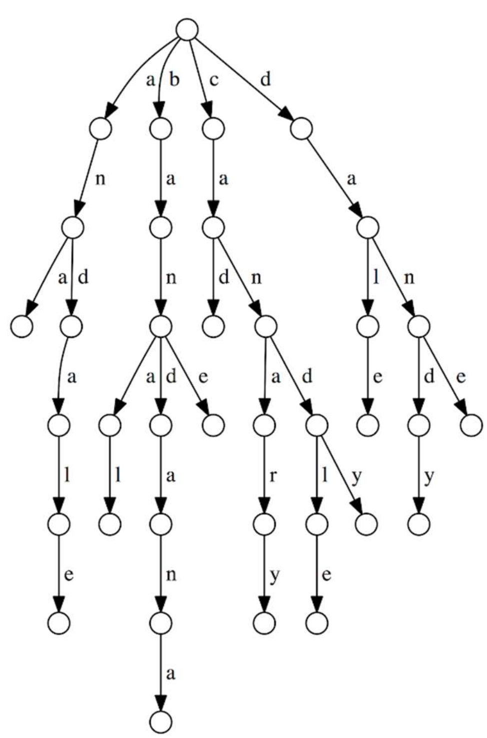
    O(|*Text*|*|*Patterns*| + |*Patterns*|)

    O(|*LongestPattern*|)

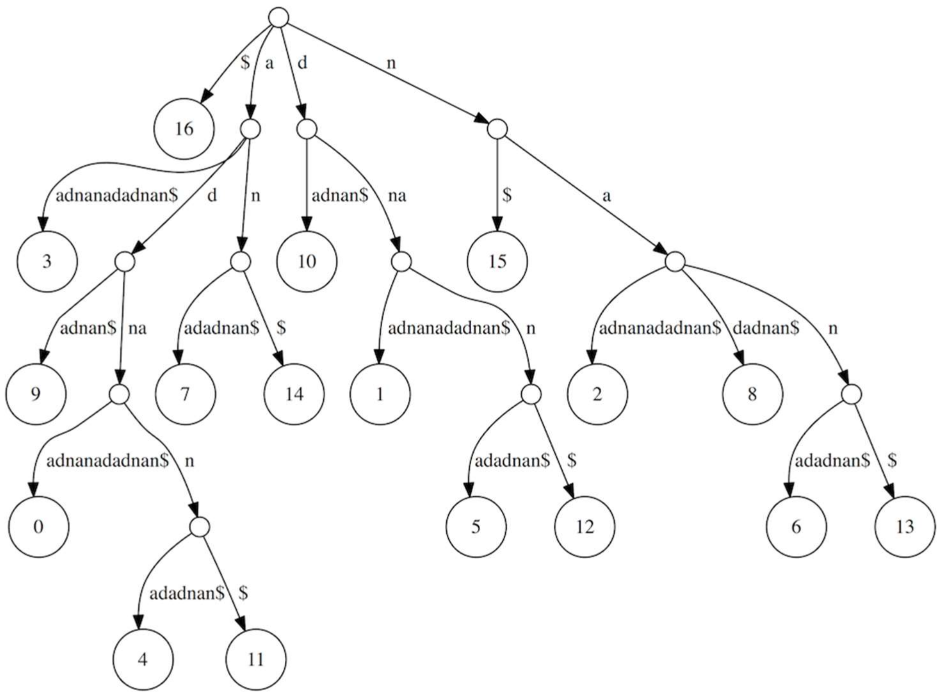
4. Which patterns could have been used to construct the following trie? (Select all that apply.)

    dale **bandana** banana

    danny **candle** **dandy**



5. Given the following suffix tree of an unknown text, identify all starting positions in the underlying text of Pattern = "na". Use 0-based indexing and enter the starting positions as space-separated integers **in increasing order**.



2 6 8 13

6. Identify all starting positions in the underlying text of Pattern = "adnan".

4 11

7. How many leaves will *SuffixTree*("TCTGAGCCCTACTGTCGAGAAATATGTATCTCGCCCCCGCAGCTT$") have?

leaves = length of genome

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Burrows-Wheeler Transform

1. Give the suffix array of "ananas$". Return your answer as a list of integers separated by spaces (e.g., 0 1 2 3 4).

   6 0 2 4 1 3 5

2. True or False: A key feature of the Burrows-Wheeler Transform is that it transforms repeats into runs.

   True

   False

3. Find the Burrows-Wheeler transform of Text = CGTTTGCTAT$.

   TT$GTCACTTG

4. If BWT(Text) = TTACA$AAGTC, what is Text?

   CACTTAAAGT$

5. Which of the following structures did we use in this chapter to decrease memory when solving the Multiple Pattern Matching Problem with the Burrows-Wheeler transform? (Select all that apply.)

   partial suffix arrays **checkpoint arrays** skew diagrams

   breakpoint graphs de Bruijn graphs

6. Say that you know that two strings of length 363 match with at most 5 mismatches, but you don't know what the strings are. What is the largest value of *k* such that we can guarantee that the two strings share a *k*-mer?

   60

Hidden Markov Models

1. Say that we are in a crooked casino where the probability of heads for the biased coin is 3/4, and we observe the following sequence of flips: *HTTTH*

   Determine whether a fair coin or a biased coin was more likely to have generated this sequence.

   Biased

   Fair

2. The definition of hidden Markov model includes which of the following objects? (Select all that apply.)

   Matrix of emission probabilities

   Matrix of transition probabilities

   Set of hidden states

   Alphabet of possible emitted symbols

   Reverse probabilities

   Viterbi graph

3. Consider an HMM with two states that emits a string of length four from an alphabet containing three symbols.

   Compute the number of edges in the Viterbi graph corresponding to this HMM (don't forget to include the source and sink nodes).

   16

4. Say that we are in a crooked casino where the probability of heads for the biased coin is 3/4.

   Given the sequence of coins π = *BFBBF* and the sequence of coin flips *x* = *HTHHH*, compute the probability Pr(*x*|π) that this sequence was generated by the given states.

   Round your answer to three decimal places.

   0.105

5. Say that we are given the following amino acid alignment (columns with *insertion rate* > θ already removed).

   M--QKCASHLE-AR

   MSNL-C-APD-LER

   MSAPNCARKYDI-R

   MS-SSCADED-IIR

   M--TKC-SKLEIDR

   What is the probability that serine (S) will be emitted from *M*(8), the 8th match state of the profile HMM corresponding to this alignment? (Write your answer as a decimal.)

   0.4